

AXI4-Stream Asynchronous FIFO with CDC Verification

Project study report - architecture, dataflow, CDC, pointers, memory, reset, and verification

1. Project Overview

The project implements a parameterized asynchronous FIFO wrapped by AXI4-Stream interfaces. The FIFO transfers packetized data between two independent clock domains while preserving ordering and TLAST packet boundaries. The write side uses **s_axis_aclk**; the read side uses **m_axis_aclk**. In the current configuration, TDATA is 8 bits, the FIFO stores 9 bits internally by adding TLAST, and the FIFO depth is 16 entries.

The top-level bridge converts AXI handshakes into FIFO write/read operations. The FIFO core then manages storage, binary read/write pointers, Gray-coded pointers, CDC synchronizers, and full/empty status.

2. Main AXI Signals

Signal	Role
s_axis_aclk	Write-domain clock (100 MHz in the testbench).
m_axis_aclk	Read-domain clock (~38.46 MHz in the testbench).
s_axis_tdata	8-bit input payload.
s_axis_tlast	Marks the final beat of a packet.
s_axis_tvalid	Source says input data is valid.
s_axis_tready	FIFO says it can accept a beat; low when full.
m_axis_tdata	8-bit output payload.
m_axis_tlast	Preserved packet-end marker.
m_axis_tvalid	FIFO says output data is available; low when empty.
m_axis_tready	Downstream receiver is ready to accept a beat.

An AXI transfer occurs only when **TVALID && TREADY** is true on the active clock edge. On the input side, this gives **winc_int**. On the output side, it gives **rinc_int**. If a transfer is stalled, the sender must hold the transaction stable.

3. Internal Data and Memory

The bridge packs each AXI beat as **{TLAST, TDATA}**. Therefore, an 8-bit AXI payload becomes a 9-bit FIFO word. With **ADDR_WIDTH = 4**, the memory contains $2^4 = 16$ entries, so the storage is conceptually **16 x 9 bits**.

The lower four bits of the binary write/read pointers select RAM addresses: **wbin[3:0]** and **rbin[3:0]**. The internal memory write is synchronous to the write clock. In this RTL model, the read output is combinational: **rdata = mem[raddr]**. The read pointer advances only after an AXI output handshake, so backpressure holds the current output entry.

4. Binary Pointers and the Extra Bit

The FIFO depth is 16, so four bits are sufficient for a RAM address. However, the read/write pointers are five bits wide: **[4:0]**. The lower four bits select one of the 16 memory locations; the extra MSB records the wrap/cycle information needed to distinguish empty from full.

For example, the write pointer progresses from **01111** to **10000**. Both values address memory location 15 and 0 respectively through their lower four bits, while the extra bit changes from 0 to 1. Thus the pointer can show that the writer has completed one full trip around the circular buffer. The five-bit pointer wraps after 32 counts.

5. Gray-Code Pointers and CDC

Binary pointers are used locally because they support normal incrementing and memory addressing. Before a pointer crosses into the other clock domain, it is converted to Gray code using **Gray = Binary XOR (Binary >> 1)**. Gray code is used because only one bit changes between adjacent pointer values, reducing the risk of an inconsistent multi-bit sample during clock-domain crossing.

The write pointer crosses into the read domain through a two-stage synchronizer: **wptr -> rq1_wptr -> rq2_wptr**. The read pointer crosses into the write domain through **rptr -> wq1_rptr -> wq2_rptr**. The first synchronizer stage may become metastable; the second provides additional resolution time and greatly reduces the chance that metastability propagates into destination logic. The synchronizers transfer pointer information, not the data payload itself.

6. FULL and EMPTY Logic

The read side determines EMPTY using the synchronized write pointer. The core compares the **next Gray-coded read pointer** against **rq2_wptr**. If they match, there are no unread FIFO entries, so **empty = 1** and the AXI output drives **m_axis_tvalid = 0**.

The write side determines FULL using the synchronized read pointer. FULL occurs when the next write pointer is one complete FIFO depth ahead of the read pointer. In this Gray-coded implementation, the full comparison matches the next write Gray pointer against the synchronized read Gray pointer with the upper two Gray bits inverted. When **wfull = 1**, **s_axis_tready = 0**, applying AXI backpressure and preventing overwrite of unread data.

Because the opposite pointer is synchronized through two flip-flops, FULL and EMPTY reflect the other clock domain with intentional CDC latency. This can temporarily make the FIFO conservative: the writer may wait a little longer to see freed space, or the reader may wait a little longer to see newly written data.

7. Reset Behavior

The design uses separate active-low asynchronous resets: **s_axis_aresetn** for the write domain and **m_axis_aresetn** for the read domain. A value of 0 asserts reset; 1 releases that domain for normal operation. Reset initializes pointer/state logic to a known condition, including **wbin = 0**, **rbin = 0**, **wfull = 0**, and **empty = 1**. The RAM array is not cleared because pointer state determines which memory entries are logically valid.

For this project, the intended operating assumption is that both clock domains are initialized together and both resets are released before normal traffic begins. Reset should be treated as flushing the logical FIFO contents; previously stored RAM bits are not guaranteed to remain valid across a one-sided or mid-transaction reset.

8. Verification Strategy and Results

The SystemVerilog testbench uses independent 10 ns and 26 ns clocks, randomized packet lengths, randomized data, randomized source idle periods, and randomized downstream backpressure. A self-checking scoreboard compares each received beat against the expected sequence, including both TDATA and TLAST. SVA assertions check AXI stability behavior, including holding TVALID and data stable when stalled, and detect unknown control signals.

Observed randomized result: 50 packets, 411 expected beats, 411 beats sent, 411 beats received, and 0 errors/mismatches. The design also met the reported Vivado setup and hold timing constraints with 0 failing endpoints in the implementation report.

9. One-Page Mental Model

Input: AXI handshake -> **winc** -> pack {TLAST, TDATA} -> write to memory at **wbin[3:0]** -> advance **wbin** -> Gray-code **wptr** -> synchronize into read domain. The read side uses the synchronized write pointer to clear EMPTY and assert **m_axis_tvalid**. When the downstream handshakes, **rinc** advances **rbin**, selecting the next memory entry. The read pointer is Gray coded and synchronized back into the write domain, where it controls FULL. FULL drives **s_axis_tready** low to prevent overflow. This closed loop safely transfers ordered AXI4-Stream data between unrelated clocks.

Key parameters: TDATA_WIDTH = 8 bits, FIFO internal word = 9 bits (TLAST + TDATA), ADDR_WIDTH = 4, FIFO depth = 16 entries, pointer width = 5 bits.